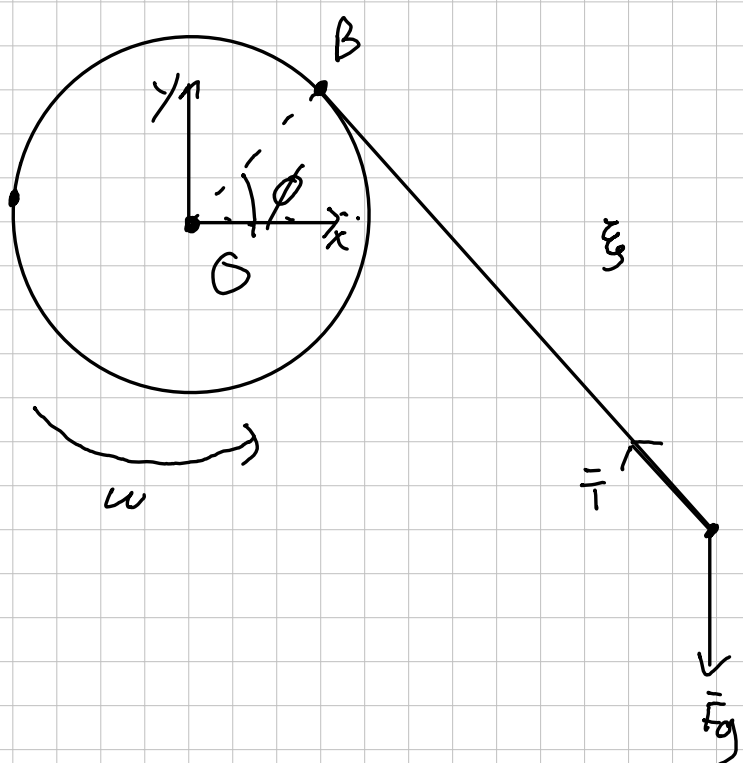




$$\hat{e}_r = \cos\phi \hat{i} + \sin\phi \hat{j}$$

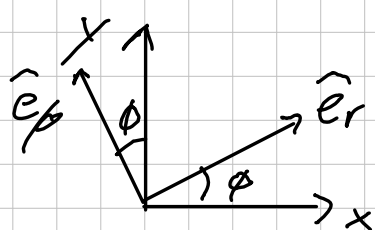
$$\dot{\hat{e}}_r = \dot{\phi} \hat{e}_\phi$$

$$\hat{e}_\phi = -\sin\phi \hat{i} + \cos\phi \hat{j}$$

$$\dot{\hat{e}}_\phi = -\dot{\phi} \hat{e}_r$$

constraint: $\xi = \xi(0) - a(\omega t - \phi)$ $\dot{\xi} = a\dot{\phi} - a\omega$

C.D.o.F. = $2 - 1 = 1$ (ϕ)



$$\hat{i} = \hat{e}_r \cos\phi - \hat{e}_\phi \sin\phi$$

$$\hat{j} = \hat{e}_r \sin\phi + \hat{e}_\phi \cos\phi$$

kinem:

$$\vec{r}_O^P = a \hat{e}_r - \xi \hat{e}_\phi$$

$$\begin{aligned} \vec{v}_O^P &= \cancel{\dot{a} \hat{e}_r} + a \dot{\phi} \hat{e}_\phi - (a \dot{\phi} - a \omega) \hat{e}_\phi + \xi \dot{\phi} \hat{e}_r \\ &= \xi \dot{\phi} \hat{e}_r + a \omega \hat{e}_\phi \end{aligned}$$

$$\begin{aligned} \vec{a}_O^P &= (a \dot{\phi} - a \omega) \dot{\phi} \hat{e}_r + \xi \ddot{\phi} \hat{e}_r - \xi \dot{\phi}^2 \hat{e}_\phi + \cancel{\dot{a} \hat{e}_\phi} + \cancel{a \ddot{\phi} \hat{e}_\phi} - a \omega \dot{\phi} \hat{e}_r \\ &= [a \dot{\phi}^2 + \xi \ddot{\phi} - 2a \omega \dot{\phi}] \hat{e}_r + \xi \dot{\phi}^2 \hat{e}_\phi \end{aligned}$$

Dynamics:

$$\vec{T} = T \hat{e}_\phi$$

$$\vec{F}_g = -mg \hat{j} = -mg (\hat{e}_r \sin\phi + \hat{e}_\phi \cos\phi)$$

$$\Sigma \vec{F} = m \vec{a} \Rightarrow$$

$$\begin{cases} \hat{e}_r: m(a \dot{\phi}^2 + \xi \ddot{\phi} - 2a \omega \dot{\phi}) = -mg \sin\phi \\ \hat{e}_\phi: m \xi \dot{\phi}^2 = T - mg \cos\phi \end{cases}$$